

# CSE221: Algorithms

## Lab-01

### Spring 2026

#### Action required for Students

##### [Google Form] CSE221 Lab Policy:

- The lab policy file can be found here - [Link](#)
- Please complete the lab policy acknowledgement form - [Link](#)

##### [Google Form] Codeforces Account Information: (Action required)

- Open an account at <https://codeforces.com/>
- Please submit your codeforces details here - [Link](#)

#### Necessary Resources [Bookmark the Link]

##### Student Drive Folder: [Link](#)

The folder contains:

- Lab Policy
- Sample Quiz Checkers
- Environment Setup for Quiz Checker
- Previous Semesters' Lab Quiz Questions

#### Assignment-01 Links:

##### codeforces Invitation Link:

<https://codeforces.com/contestInvitation/d920b97134683f693e4c44cb493e8a192b4b705d>

Alternative Link: <https://codeforces.com/contests/670025>

[The alternative link will **only work** after you have successfully registered using the first link.]

**Deadline:** Feb 15, 2026 2:00 PM

After the deadline ends, you can submit your solution by entering the contest as a Virtual Participant for practice purposes. **However, please note, submissions after the deadline are not accepted for grading.**

#### **Learning Outcome of this Lab:**

- Understand the basics of Online Judge platforms.
- Become familiar with standard Input/Output (I/O) formats used in Online judge
- Learn about common judge verdicts, including Accepted (AC), Wrong Answer (WA), Time Limit Exceeded (TLE), and Runtime Error (RE).
  - For a **1-second** time limit, most OJs allow approximately  **$10^8$  operations.**
  - For python select PyPy over Python if available, as PyPy is faster. However, PyPy consumes more memory than Python.
- Review  $O(n^2)$  sorting algorithms which have already been covered in the previous courses such as Bubble Sort, Selection Sort, and Insertion Sort.
  - Learn to sort with multiple parameter
  - Understand the use case of stable sort
- Explore Fast I/O techniques (**self-study**):
  - **Python:** Using `sys.stdin` for faster input handling.
  - **C++:** Using `ios_base::sync_with_stdio(false);` and `cin.tie(nullptr)`
  - **Java:** Using `BufferedReader` for efficient input reading.
  - The fast i/o can reduce program execution time on Online Judge platforms.
  - **Fast I/O Example:** [Python](#), [C++](#), [Java](#) [click Expand to view the code snippet]